# ANTI-FT: TOWARDS PRACTICAL DEEP LEAKAGE FROM GRADIENTS

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## ABSTRACT

Federated learning is usually regarded as a privacy-preserving training paradigm for it enables multiple clients to participate in a training task without sharing their private data. However, recent studies revealed that a malicious server can still recover private data from the victim clients based on the shared gradients via deep leakage from gradients (DLG). Currently, almost all DLG attacks are designed based on the average loss, leading to a significant decrease in attack efficiency when the batch size is greater than 1. In this paper, we revisit DLG attacks from the perspective of the loss function. We reveal that not all samples in the target batch are equally susceptible to DLG attacks: the sample with the highest loss value tends to be easily recovered by DLG attacks. Based on these observations, we propose a simple yet effective DLG method under practical FL settings. Specifically, the adversaries can enhance the effectiveness of DLG by perturbing the global model through finetuning it with a few mislabeled samples (dubbed 'Anti-FT'). Extensive experiments are conducted on benchmark datasets, which verify the effectiveness of our method and its resistance to potential defenses. The codes are available at https://github.com/zlh-thu/anti-finetune.

***Index Terms*—** Deep Leakage from Gradients, Federated Learning, Trustworthy ML, AI Security.

## 1. INTRODUCTION

Deep neural networks (DNNs) have been adopted in many applications, such as facial recognition [1, 2] and speech recognition [3, 4, 5]. Generally, training a well-performed DNN requires a large number of training samples that may be dispersed among clients. How to train a global model without directly collecting those samples is of great significance since they may contain sensitive information and the collection is prohibited by some regulations ($e.g.$, GDPR [6]).

To address the aforementioned problem, federated learning (FL) [7, 8] has rapidly emerged as a new training paradigm where clients can collaboratively train a global model without uploading their data. Specifically, each client learns from its local samples and shares the computed gradient or updated model with the server, which helps to mitigate privacy risks of the local data. The server will aggregate all shared components, based on which to update the global model [8, 9].

However, recent studies [10, 11, 12, 13, 14, 15, 16] have shown that transmitting gradients alone is not as secure as previously believed. A malicious server can reconstruct local images by minimizing the distance between the gradients of the recovered image and those uploaded by the victim client. This attack, known as deep leakage from gradients (DLG), poses a significant privacy risk.

Generally, the settings of existing DLG attacks can be classified into two main categories. In the first setting, 'an honest-but-curious' server has access to the global model and the clients' shared local model parameters but is not permitted to modify them [10, 11, 12, 13, 14]. In the second setting, a malicious server can manipulate both the global model and the clients' shared model parameters [15, 16]. By modifying the global model before distributing it to clients, the server can enhance its ability to extract training data, posing a greater threat to client privacy.

However, we notice that most existing DLG attack methods can only recover local samples when the batch size is small [10, 12, 13, 14, 17] or have to be one [11], even under additional requirements. For example, under the first setting, even the most advanced method ($i.e.$, GradInversion (GI) [13]) has only demonstrated success with batch sizes up to 48, with an additional requirement that all samples in the batch belong to different categories. This condition is rarely met in practice. Besides, GI requires the target model to have batch normalization (BN) layers since it leverages batch-wise mean and variance estimates as priors for optimization. For

This work is supported by the National Key Research and Development Project of China (Grant No. 2023YFF0905502), the National Natural Science Foundation of China(Grant No. 92467204 and 62472249), and the Shenzhen Science and Technology Program (Grant No. JCYJ20220818101014030 and KJZD20240903102300001).

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the second type of DLG methods [15, 16], they allow image inversion for larger batch sizes but greatly sacrifice attack stealthiness. Specifically, Fishing [15] directly sets the chosen parameters in the fully connected layer to zero to increase the probability that easily recoverable samples appear in the target batch. Loki [16] strengthens the attack by injecting additional model structures. These obvious modifications make it easier for the client to detect the DLG attack. Accordingly, an intriguing and important question arises: *Shall we design a second type of DLG attack that is both effective and stealthy?*

To answer the aforementioned question, we first revisit the success mechanisms and conditions of the first type of DLG attacks. We reveal that these DLG attacks tend to recover 'hard' samples with the highest loss value in their data batch. This is mostly because the information from these samples occupies a dominant position in the average process. Based on this understanding, we propose Anti-FT, a simple, undetectable, yet effective DLG attack method (in the second setting). Specifically, before delivering the global model, the server finetunes it with a small amount of poisoned public data. By finetuning the global model, hard samples are more likely to appear in the target batch, instead of all samples exhibiting similar loss values without any distinct outliers. The presence of hard samples significantly facilitates image inversion, thereby enhancing the effectiveness of DLG attacks.

In conclusion, the main contributions of this paper are three-fold. **(1)** We reveal that not all samples are equally susceptible to DLG attacks. Specifically, DLG attacks tend to recover 'hard' samples having the highest loss value in their data batch. **(2)** We revisit DLG attacks from the perspective of the loss function and propose a simple yet effective DLG method ($i.e.$, Anti-FT). **(3)** We conduct extensive experiments on benchmark datasets, verifying the effectiveness of our method and its resistance to potential defenses.

## 2. MOTIVATION: NOT ALL SAMPLES ARE EQUALLY SUSCEPTIBLE TO DLG ATTACKS

In this section, we revisit the major limitations of DLG attacks by analyzing samples that can be successfully recovered. We use the IG attack [12] as an example. Specifically, we perform the IG attack with batch sizes ($i.e.$, $|\mathcal{B}| \in \{4, 8\}$) on the CIFAR-10 [18] dataset for 300 and 500 times, respectively. We train a ResNet-18 [19] model on CIFAR-10 for 100 epochs as the target model $f$. During each attack, we randomly select the target batch $\mathcal{B} = \{(\boldsymbol{x}_i, y_i) | 1 \leq i \leq |\mathcal{B}|\}$. The recovered batch is denoted as $\mathcal{B}^* = \{(\boldsymbol{x}_i^*, y_i^*) | 1 \leq i \leq |\mathcal{B}|\}$. We use the Peak Signal-to-Noise Ratio (PSNR) as the quantitative evaluation. The higher the PSNR, the better the recovery result. We denote the recovered result of $\boldsymbol{x}_i$ as $\boldsymbol{x}_i^*$. To further analyze the behavior of DLG attacks, we also calculate the loss value of every sample in the target batch. We denote the sample with the highest loss value in each $\mathcal{B}$ as $\boldsymbol{x}_M$ and its recovery result is denoted as $\boldsymbol{x}_M^*$.

As shown in Table 1 and Figure 1, we find that it is difficult for the attack to recover all samples in the batch, and the PSNR of the sample with the highest loss is higher than the mean PSNR value of the entire batch. Specifically, *compared to other samples, DLG is more effective at recovering 'hard' samples* having a large loss value. These results reveal that not all samples are equally susceptible to DLG attacks.

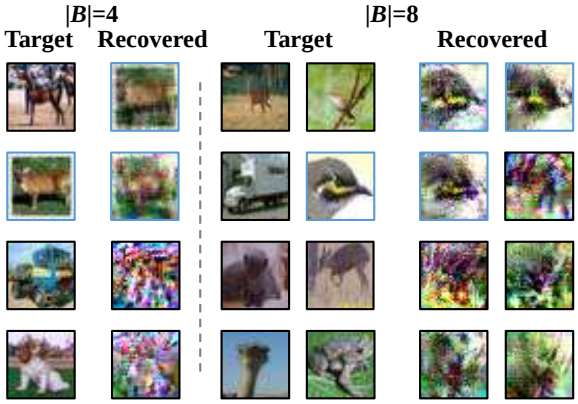


**Fig. 1**. The results of the IG attack with batch size $|\mathcal{B}| \in \{4, 8\}$. The samples with the highest loss values in the target batches and their correspondingly recovered samples are both indicated in the blue box.

**Table 1**. The results of IG attack with batch size $|\mathcal{B}| \in \{4, 8\}$ on the CIFAR-10 dataset.

| Batch Size | Attack Times | Mean value of PSNR($\boldsymbol{x}_i, \boldsymbol{x}_i^*$) | PSNR($\boldsymbol{x}_M, \boldsymbol{x}_M^*$) |
|---|---|---|---|
| $\lvert\mathcal{B}\rvert = 4$ | 300 | 10.30 | 17.40 |
| $\lvert\mathcal{B}\rvert = 8$ | 500 | 10.24 | 15.91 |

## 3. METHODOLOGY

### 3.1. Problem Formulation

**The Main Pipeline of FL.** Considering an $N$-classification task, the clients and the server will conduct global training for $\mathcal{T}$ iterations. At the $t$-th iteration, after receiving the global model $f_t$, the $k$-th client will train $f_t$ with a batch of data $\mathcal{B}$ of the local dataset $\mathcal{D}_k$ and then get the local model $f_t^k$ as:

$$f_t^k = f_t - \eta \nabla_{f_t} \mathcal{L}(f_t(\mathcal{B})), \tag{1}$$

where $\eta$ is the learning rate, $\mathcal{L}$ denotes the loss function ($i.e.$, cross-entropy). The server then updates the global model as:

$$f_{t+1} = \sum_{k}^{K} \frac{n_k}{n} f_t^k, \tag{2}$$

where $K$ is the total number of involved clients, $n_k$ is the number of training data in local dataset $\mathcal{D}_k$ and $n = \sum_k^K n_k$.

**The Main Pipeline of DLG.** The DLG adversaries aim to recover images based on the gradients of the global model. Taking the Inverting Gradients (IG) attack [12] as an example of current attack process, the adversary can easily infer the

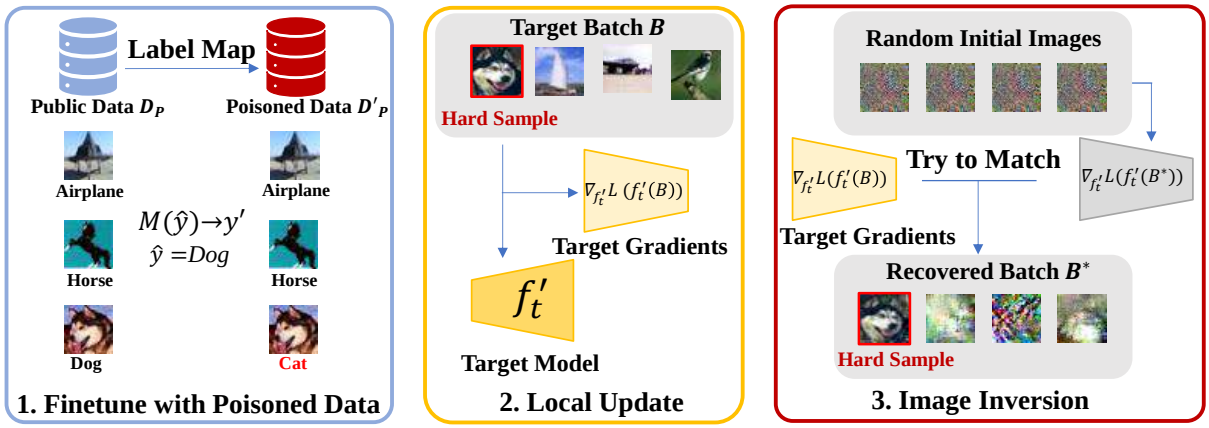


**Fig. 2**. The main pipeline of deep leakage from gradients with our Anti-FT. Step 1. Finetune with Poisoned Data: Finetuning the global model using a small amount of poisoned public data to increase the likelihood of hard samples appearing in the target batch; Step 2. Local Update: Clients perform local updates on their data using the finetuned global model, generating gradients that carry information about hard samples; Step 3. Image Inversion: The adversary attempts to recover images from client-side gradients, leveraging the dominance of hard samples in the gradient information.

gradients of $f_t$ according to Equation (1) by:

$$\nabla_{f_t}\mathcal{L}(f_t(\mathcal{B})) = \frac{1}{\eta}(f_t - f_t^k). \tag{3}$$

The adversary can optimize a batch of randomly initialized images, $\mathcal{B}^* \sim \mathcal{N}(0,1)$, to approximate the target batch $\mathcal{B}$ by minimizing the difference between their gradients. Specifically, the attack is performed by minimizing the cosine distance between the gradients of $\mathcal{B}^*$ and $\mathcal{B}$, formulated as:

$$\arg\min_{\mathcal{B}^*} 1 - \frac{\langle \nabla_{f_t}\mathcal{L}(f_t(\mathcal{B})), \nabla_{f_t}\mathcal{L}(f_t(\mathcal{B}^*))\rangle}{||\nabla_{f_t}\mathcal{L}(f_t(\mathcal{B}))||||\nabla_{f_t}\mathcal{L}(f_t(\mathcal{B}^*))||} + \alpha\cdot\mathcal{V}(\mathcal{B}^*), \tag{4}$$

where $\mathcal{V}(\cdot)$ represents the total variation [20], and $\alpha$ is its corresponding hyper-parameter.

**Threat Model.** Following existing works [15, 16], we consider a setting where a malicious server is the adversary. The server can actively manipulate the current global model $f_t$ and the clients' shared local model parameters $f_t^k$, a scenario referred to as the server-side model manipulation setting. Additionally, we assume the server has a small number of public data. Similar assumption has been used extensively in other leakage attacks [21, 22].

### 3.2. Deep Leakage from Gradients with Anti-FT

Based on the insights from Section 2, samples with the highest loss values in the target batch (*i.e.*, hard samples) are more vulnerable to gradient leakage attacks. Therefore, the server can enhance the DLG attack by increasing the likelihood of hard samples appearing in the target batch $\mathcal{B}$. To achieve this, we propose the Anti-FT method to perturb the global model with a small amount of poisoned public data. By finetuning the global model with poisoned public data, hard samples are more likely to appear in the target batch, instead of all samples exhibiting similar loss values without any distinct outliers. As shown in Figure 2, Anti-FT consists of three steps.

**1. Finetune with Poisoned Data.** Let $M : \mathcal{Y} \rightarrow \mathcal{Y}$ denote a function that maps a target label $\hat{y} \in [\mathcal{N}] = \{1, 2, ..., N\}$ randomly to another category $y' \in [\mathcal{M}]/[\hat{y}] = \{1, ..., \hat{y}-1, \hat{y}+1, ..., N\}$. The server can generate a poisoned version of the public dataset $\mathcal{D}'_P = \{(\boldsymbol{x}, y')|y' = M(\hat{y}), (\boldsymbol{x}, \hat{y}) \in \mathcal{D}_P)\}$. Before delivering the latest global model $f_t$ to the target clients, the server can perturb $f_t$ by finetuning it with $\mathcal{D}'_P$ for $E$ epochs and get the perturbed global model $f'_t$.

**2. Local Update.** Let the $T$-th client denote a target client. After receiving the perturbed global model $f'_t$, the local update can be conducted as:

$$f_t'^T = f'_t - \eta\nabla_{f'_t}\mathcal{L}(f'_t(\mathcal{B})), \tag{5}$$

where $\mathcal{B}$ is the target batch and $f_t'^T$ is the latest local model of the target client.

**3. Image Inversion.** After the server receives $f_t'^T$, the gradient of $f'_t$ can be easily inferred as:

$$\nabla_{f'_t}\mathcal{L}(f'_t(\mathcal{B})) = \frac{1}{\eta}(f'_t - f_t'^T). \tag{6}$$

Thus, with the perturbed global model $f'_t$, DLG attack may recover the hard samples in target batch $\mathcal{B}$ via:

$$\arg\min_{\mathcal{B}^*} 1 - \frac{\langle \nabla_{f'_t}\mathcal{L}(f'_t(\mathcal{B})), \nabla_{f'_t}\mathcal{L}(f'_t(\mathcal{B}^*))\rangle}{||\nabla_{f'_t}\mathcal{L}(f'_t(\mathcal{B}))||||\nabla_{f'_t}\mathcal{L}(f'_t(\mathcal{B}^*))||} + \alpha\cdot\mathcal{V}(\mathcal{B}^*), \tag{7}$$

where $\mathcal{V}(\cdot)$ is total variation [20] and $\alpha$ is a hyper-parameter.

## 4. EXPERIMENTS

### 4.1. Main Settings

**Dataset and Model Selection.** We evaluate Anti-FT on CIFAR-10 [18] and a 20-class subset of ImageNet [24] which includes 10,000 training samples (500 per class). Following prior work [12, 13, 14, 15, 16], we use ResNet-18 [19] as

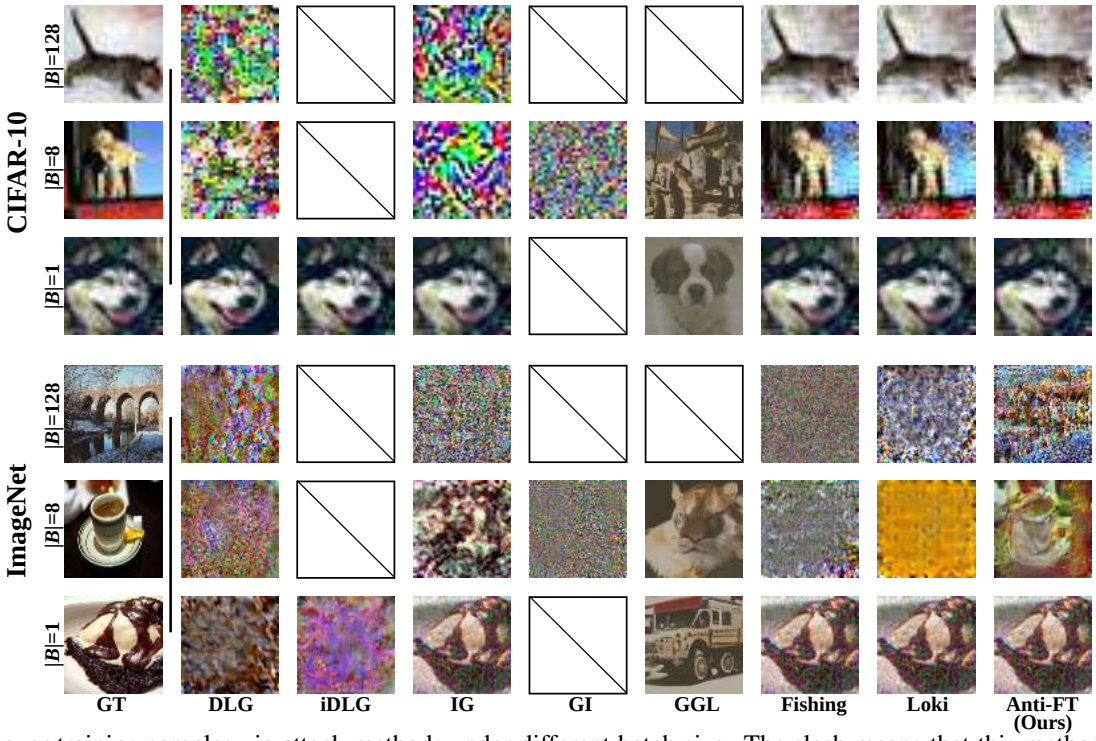


**Fig. 3**. Recover training samples via attack methods under different batch size. The slash means that this method can not be performed in this scenario due to limitations of batch size or model architecture. iDLG can infer the ground-truth label from gradients only when $|\mathcal{B}| = 1$. Both GI and GGL operate under the assumption that there are no duplicate labels within the target batch, implying that the batch size must be smaller than the number of classes. Additionally, GI assumes the presence of BatchNorm layers [23], which prevents it from being applied when $|\mathcal{B}| = 1$. (GT denotes the ground truth images.)

the global model in FL. Since the BN layers are incompatible with $\mathcal{B} = 1$, we remove the BN layers from all model architectures in experiments with $\mathcal{B} = 1$.

**Settings for FL.** We assume FL involves 8 clients ($K = 8$), each using the SGD optimizer with a learning rate of 0.01 ($\eta = 0.01$). To evaluate Anti-FT, we conduct FL training with batch sizes $|\mathcal{B}| \in \{1, 8, 128\}$. Each client receives an equal number of training samples: 6250 for CIFAR-10 ($n_k = 6,250$) and 1250 for ImageNet ($n_k = 1,250$). Clients perform one local training step with one batch and share the updated model after each step. The training process contains 200 global epochs ($E_g = 200$) where one global epoch is defined as all local data being used once, $i.e.$, $\mathcal{T} = \frac{E_g \times n_k}{|\mathcal{B}|}$.

**Attack Setup and Baseline Methods.** We assume the 1st client is the attack target, with the adversary conducting attacks only in the last 100 global epochs. In the first 100 epochs, the server updates the global model with Equation (2). In the last 100 global epochs, the adversary will perform the attack for every training iteration, that number of attacks can be noted as $100 \times \frac{n_k}{|\mathcal{B}|}$. We randomly sample 2,000 and 1,000 samples from the testing set as the public dataset $\mathcal{D}_P$ for CIFAR-10 and ImageNet respectively. For both datasets, the server randomly chooses a target label and conducts Anti-FT for 5 epochs ($E = 5$) with the poisoned dataset $\mathcal{D}'_P$. To comprehensively evaluate the effectiveness of the proposed method, we consider baseline attacks under two settings: (1) the server-side model manipulation setting and (2) the honest-but-curious server setting. For the server-side model manipulation setting, we compare our method with Fishing [15] and Loki [16]. We implement these methods following the authors' released code repositories [11, 10, 12, 14, 15, 16]. For the honest-but-curious server setting, we compare our method with state-of-the-art attacks, including Deep Leakage from Gradients (DLG) [10], improved DLG (iDLG) [11], Inverting Gradients (IG) [12], GradInversion (GI) [13], and Generative Gradient Leakage (GGL) [14].

**Evaluation Metrics.** We use the peak signal-to-noise ratio (PSNR) and the structural similarity index (SSIM)[25] as the quantitative evaluation. Higher PSNR and SSIM values indicate better attack performance. We take the reconstructed images with PSNR values greater than 20 as successful attacks for CIFAR-10 and 15 for ImageNet. To evaluate the stealthiness of the attack methods, we also assess the accuracy of the global model, as abnormal accuracy, $e.g.$, an unusually low value, may reveal the presence of an attack. Among all attack methods, the best result is highlighted in boldface.

### 4.2. Main Results

For server-side model manipulation methods, as shown in Table 4-5 and Figure 3, Anti-FT achieves the best image inversion results in almost all cases. Unlike Loki, **Anti-FT does not require modifications to the global model's architecture**, making the attack more difficult for clients to detect. Additionally, **Anti-FT has minimal impact on the global model's accuracy**, further enhancing its stealthiness.

**Table 2**. Results of methods under the honest-but-curious server setting on the CIFAR-10 dataset.

| $\|\mathcal{B}\|$ | DLG | | | iDLG | | | IG | | | GI | | | GGL | | | Anti-FT | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | ACC | PSNR | SSIM | ACC | PSNR | SSIM | ACC | PSNR | SSIM | ACC | PSNR | SSIM | ACC | PSNR | SSIM | ACC | PSNR | SSIM |
| 1 | **78.47%** | 9.64 | 0.30 | **78.47%** | 9.41 | 0.29 | **78.47%** | 19.82 | 0.87 | – | – | – | **78.47%** | 12.35 | -0.11 | 76.08% | **19.90** | **0.88** |
| 8 | **78.54%** | 10.58 | 0.30 | – | – | – | **78.54%** | 8.61 | 0.31 | 32.19% | 7.92 | 0.29 | **78.54%** | 11.78 | -0.10 | 77.69% | **12.43** | **0.46** |
| 128 | 66.74% | 5.32 | 0.02 | – | – | – | 66.74% | 11.81 | 0.37 | – | – | – | – | – | – | **71.74%** | **12.35** | **0.40** |

**Table 3**. Results of methods under the honest-but-curious server setting on the ImageNet dataset.

| $\|\mathcal{B}\|$ | DLG | | | iDLG | | | IG | | | GI | | | GGL | | | Anti-FT | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | ACC | PSNR | SSIM | ACC | PSNR | SSIM | ACC | PSNR | SSIM | ACC | PSNR | SSIM | ACC | PSNR | SSIM | ACC | PSNR | SSIM |
| 1 | 50.90% | 9.71 | 0.16 | 50.90% | 9.83 | 0.18 | 50.90% | 12.03 | 0.45 | – | – | – | 50.90% | 10.83 | 0.05 | **51.70%** | **12.50** | **0.50** |
| 8 | 43.50% | 10.17 | 0.07 | – | – | – | 43.50% | 8.17 | 0.08 | 43.50% | 7.92 | 0.08 | 43.50% | **11.68** | 0.13 | **43.80%** | 11.47 | **0.24** |
| 128 | **28.10%** | 9.56 | 0.09 | – | – | – | **28.10%** | 8.44 | 0.08 | – | – | – | – | – | – | 27.60% | **9.71** | **0.12** |

**Table 4**. Results of methods under the server-side model manipulation setting on the CIFAR-10 dataset.

| $\|\mathcal{B}\|$ | Fishing | | | Loki | | | Anti-FT | | |
|---|---|---|---|---|---|---|---|---|---|
| | ACC | PSNR | SSIM | ACC | PSNR | SSIM | ACC | PSNR | SSIM |
| 1 | 59.77% | 19.87 | 0.90 | 70.92% | 19.56 | **0.91** | **76.08%** | **19.90** | 0.88 |
| 8 | 57.06% | 11.15 | 0.39 | 72.17% | **12.66** | 0.44 | **77.69%** | 12.43 | **0.46** |
| 128 | 64.92% | 12.04 | **0.40** | 66.20% | 12.14 | 0.34 | **71.74%** | **12.35** | **0.40** |

**Table 5**. Results of methods under the server-side model manipulation setting on the ImageNet dataset.

| $\|\mathcal{B}\|$ | Fishing | | | Loki | | | Anti-FT | | |
|---|---|---|---|---|---|---|---|---|---|
| | ACC | PSNR | SSIM | ACC | PSNR | SSIM | ACC | PSNR | SSIM |
| 1 | 47.50% | 11.50 | 0.48 | 49.90% | 12.10 | 0.47 | **51.70%** | **12.50** | **0.50** |
| 8 | 37.50% | 11.20 | 0.09 | 40.25% | 11.05 | **0.25** | **43.80%** | **11.47** | 0.24 |
| 128 | 10.60% | 9.00 | **0.14** | 20.66% | 9.60 | 0.08 | **27.60%** | **9.71** | 0.12 |

**Table 6**. Number of recovered images (PSNR>20) from the target client for 100 global training epochs on CIFAR-10.

| $\mathcal{B}$ | Attack times | DLG | iDLG | IG | GI | GGL | Fishing | Loki | Anti-FT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 625,000 | 687 | 801 | 3,122 | – | 2,301 | 3,248 | **3,496** | 3,101 |
| 8 | 78,125 | 0 | – | 156 | 0 | 20 | 280 | 398 | **433** |
| 128 | 4,883 | 0 | – | 0 | – | – | 10 | 14 | **17** |

**Table 7**. Number of recovered images (PSNR>15) from the target client for 100 global training epochs on ImageNet.

| $\mathcal{B}$ | Attack times | DLG | iDLG | IG | GI | GGL | Fishing | Loki | Anti-FT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 125,000 | 0 | 0 | 114 | – | 35 | 117 | 122 | **128** |
| 8 | 15,625 | 0 | – | 0 | 0 | 10 | 8 | **12** | 9 |
| 128 | 977 | 0 | – | 0 | – | – | 1 | 0 | **2** |

For a comprehensive comparison, we also conduct attacks under the honest-but-curious server setting. As shown in Table 2-3 and Figure 3, among the baseline methods, only DLG and IG can consistently perform leakage attack in all cases without any limitations. Other baseline methods can not be applied in specific scenarios due to the batch size or model architecture limitations. For both RSNR ans SSIM results, our method achieves the best attack performance in almost all cases. For PSNR results, the only case where our method ranks second in image inversion performance is when $|\mathcal{B}| = 8$ on the ImageNet dataset, where GGL achieves the best result. However, as shown in Figure 3, GGL often causes a mismatch between the ground-truth image and the recovered image. This may be because GGL relies on a pre-trained generative adversarial network (GAN) [26] to assist in image inversion, and the prior knowledge embedded in the pre-trained model does not align with the target client's local data. In terms of model accuracy, Anti-FT maintains performance comparable to baseline methods. To better reflect the overall attack effectiveness, we set a threshold of successful reconstruction that PSNR$>$ 20 for CIFAR-10 samples and PSNR$>$ 15 for the ImageNet samples. As shown in Table 6-7, Anti-FT recovers a considerable number of samples.

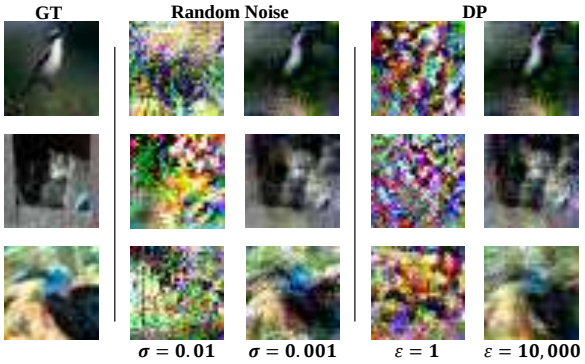


**Fig. 4**. The resistance to potential defenses.

**Table 8**. The Resistance to potential defenses.

| Defense | No Defense | Random Noise | | DP | |
|---|---|---|---|---|---|
| | | $\sigma = 0.01$ | $\sigma = 0.001$ | $\epsilon = 1$ | $\epsilon = 10,000$ |
| ACC | 71.74% | 10.00% | 57.18% | 10.00% | 40.58% |
| PSNR | 12.35 | 7.68 | 9.90 | 7.37 | 10.21 |

### 4.3. The Resistance to Potential Defenses

We conduct the experiments on CIFAR-10 with $|\mathcal{B}| = 128$ as a case to study the resistance to potential defenses. Following the existing works [10, 14], the defender injects Gaussian noise with $noise \sim \mathcal{N}(0, \sigma^2 \boldsymbol{I})$ and differential privacy (DP) [27, 28, 29, 30] noise where the privacy budget is denoted by $\epsilon$ to protect the shared model. As shown in Table 8 and Figure 4, when more noise is injected ($\sigma = 0.01$ and $\epsilon = 1$), the defender can prevent Anti-FT from achieving high-quality reconstruction. However, the model accuracy drops sharply to 10.00%. When the defender injects less noise to improve the model accuracy ($\sigma = 0.001$ and $\epsilon = 10,000$), Anti-FT remains effective, while the model accuracy either remains low (10.00%) or improves only slightly (40.58%). **This suggests that the defender may have to significantly sacrifice model accuracy to effectively defend against Anti-FT.**

## 5. CONCLUSION

In this paper, we revealed that not all samples in the target batch are equally vulnerable to DLG attacks and the sample with the highest loss value tends to be more easily recovered. Based on these findings, we proposed to enhance DLG by finetuning the model with a few mislabeled samples (dubbed 'Anti-FT'). Extensive experiments on benchmark datasets verified the effectiveness and robustness of our method.

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